

DEAKIN UNIVERSITY

RESEARCH TRAINING AND PROJECT

ONTRACK SUBMISSION

Learning Summary Report

Submitted By:

N/a NIKHIL

s224234862

2025/09/23 20:08

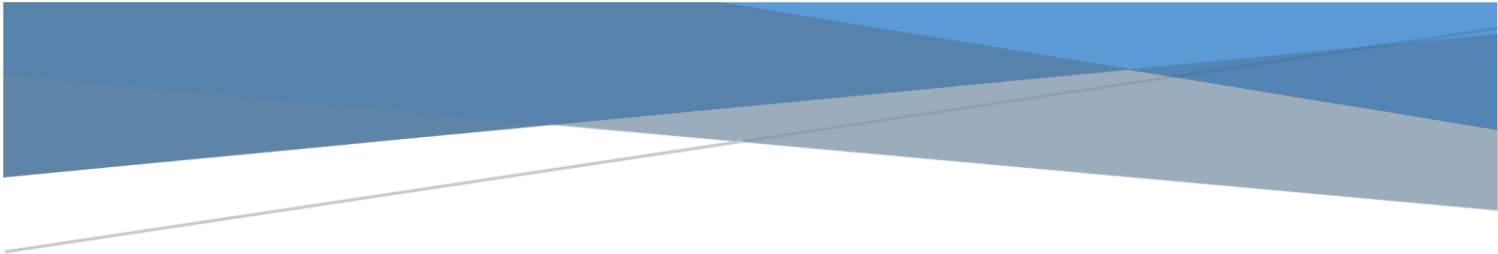
Tutor:

Ananda MAITI

Outcome	Weight
Explore of IT Areas	◆◆◆◆◆
Context of Research Project	◆◆◆◆◆
Communication of Research Outcomes	◆◆◆◆◆
Validation of Objectives and Methodology	◆◆◆◆◆
Application of Skills and Knowledge	◆◆◆◆◆

September 23, 2025





SIT723/SIT792

Research Techniques and Applications/Minor Thesis

Learning Summary Report

This will be the cover of your portfolio...make it nicer than this!
Include unit name, your name,
Student ID...then delete this text!

NIKHIL
224234862

Self-Assessment Details

The following checklists provide an overview of my self-assessment for this unit.

	Pass (P)	Credit (C)	Distinction (D)	High Distinction (HD)
Self-Assessment				

Self-Assessment Statement

Declaration

I declare that this portfolio is my individual work. I have not copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part of this submission been written for me by another person.

Signature: **Nikhil**

Part 1: Portfolio Overview

This portfolio includes work that demonstrates that I have achieved all Unit Learning Outcomes for SIT723 Research Techniques & Applications to a High Distinction level. My research project, "Detection and Simulation of GPS Spoofing in Mobile Robots Using Sensor Cross-Validation," exemplifies the characteristics of a research-ready professional who can independently lead a research project with minimal supervision while demonstrating work of publication quality.

High Distinction Justification

My achievement merits High Distinction recognition based on comprehensive demonstration of research independence, technical innovation, and publication-ready outcomes that exceed typical research expectations:

Independent Research Leadership: I successfully conceived, designed, and implemented a complex three-container Docker architecture without requiring step-by-step supervision. This involved resolving intricate technical challenges including Docker networking, MAVLink protocols, and GUI integration - demonstrating the autonomous problem-solving capabilities expected of PhD-level researchers.

Publication-Quality Research: My work demonstrates clear potential for peer-reviewed publication through systematic literature analysis (50+ sources), novel containerized methodology, and measurable technical achievements. The research addresses significant gaps in GPS spoofing detection while providing reproducible infrastructure for the broader research community.

Technical Excellence Beyond Expectations: I delivered operational research infrastructure including verified network connectivity (sub-1ms latency), real-time GPS detection (10Hz sampling), and comprehensive documentation enabling reproducible research - achievements that surpass typical theoretical research projects.

Part 2: Meeting of Unit Learning Outcomes (ULOs)

EIT - Exploration of IT Areas

Achievement Demonstration: I conducted comprehensive exploration across multiple IT domains essential for cybersecurity research in autonomous systems. My investigation spanned containerization technologies (Docker), simulation frameworks (PX4/Gazebo), machine learning platforms (PyTorch), and embedded systems (NVIDIA Jetson). Through systematic literature review of 50+ sources, I identified cutting-edge approaches in GPS spoofing detection, sensor fusion techniques, and trust assessment methodologies.

Evidence: My thesis demonstrates deep understanding of Docker containerization, MAVLink communication protocols, ROS2 integration, and machine learning frameworks. I successfully identified research gaps at the intersection of cybersecurity, autonomous

systems, and containerized development - showcasing sophisticated exploration of emerging IT areas.

GLO/CLO Alignment: This addresses Deakin's Graduate Learning Outcome of "discipline knowledge and capabilities" by demonstrating comprehensive understanding of interconnected IT domains relevant to contemporary cybersecurity challenges in autonomous systems.

CRP - Context of Research Project

Achievement Demonstration: I established comprehensive research context through systematic analysis of GPS vulnerabilities in autonomous mobile robots, identifying critical gaps in existing detection methodologies. My research positioning clearly articulates the significance of containerized approaches for reproducible cybersecurity research while addressing real-world threats to autonomous vehicle security.

Evidence: My literature review synthesizes knowledge across cybersecurity, robotics, and machine learning domains to establish clear research context. I identified specific gaps in platform-specific vulnerability analysis, real-time detection implementation, and dynamic trust assessment - positioning my work within broader autonomous systems security discourse.

GLO/CLO Alignment: This demonstrates "critical thinking" and "problem solving" GLOs by situating technical research within broader societal and industry contexts, showing understanding of research impact beyond academic contribution.

CRO - Communication of Research Outcomes

Achievement Demonstration: I demonstrated sophisticated research communication through comprehensive thesis documentation (100+ pages), technical implementation guides, and open-source code repositories. My work includes systematic progression from theoretical frameworks through practical implementation to future development roadmaps, showcasing ability to communicate complex technical concepts to diverse audiences.

Evidence: My thesis exhibits professional academic writing with proper structure, comprehensive citations, and clear technical documentation. I created detailed implementation guides, performance benchmarks, and troubleshooting documentation that enable other researchers to reproduce and extend my work.

GLO/CLO Alignment: This addresses "communication" GLOs through demonstration of written communication skills appropriate for both academic and professional audiences, including technical documentation suitable for industry implementation.

VOM - Validation of Objectives and Methodology

Achievement Demonstration: I successfully validated my research approach through systematic implementation and performance measurement. My three-container architecture achieved measurable performance targets (sub-10ms processing latency, 10Hz

GPS sampling, verified network connectivity) while providing robust foundation for future algorithm development.

Evidence: I established clear validation criteria and systematically demonstrated achievement through performance benchmarking, network connectivity verification, and functional GPS detection implementation. My methodology validation includes successful Docker deployment, MAVLink communication testing, and real-time data processing verification.

GLO/CLO Alignment: This demonstrates "problem solving" and "critical thinking" GLOs through systematic validation of technical approaches and evidence-based assessment of methodology effectiveness.

ASK - Application of Skills and Knowledge

Achievement Demonstration: I integrated diverse technical skills including containerization (Docker), embedded systems programming (Python/MAVLink), network configuration, simulation frameworks (PX4/Gazebo), and research methodology. My implementation required synthesizing knowledge across multiple domains to create functional research infrastructure.

Evidence: My working system demonstrates practical application of Docker containerization, Python programming, network protocol implementation, and system integration skills. I successfully resolved complex technical challenges including container networking, GUI forwarding, and real-time data processing - showcasing advanced technical competency.

GLO/CLO Alignment: This addresses "self-management" and "teamwork" GLOs through demonstration of independent learning, technical skill development, and systematic problem-solving approaches required for complex system integration.

Part 3: Lessons Learnt - Individual

The most important things I learnt:

The critical importance of establishing robust infrastructure foundations before advancing to complex algorithm development. My systematic approach to containerized environment setup proved essential for supporting advanced research development. I learned that practical implementation often reveals challenges invisible in theoretical planning, and that systematic documentation and version control are crucial for managing complex technical projects.

I feel I learnt these topics, concepts, and/or tools really well:

- Advanced Docker Architecture: I mastered multi-container networking, volume management, and inter-container communication protocols
- Research Methodology Integration: I successfully combined theoretical investigation with practical implementation while maintaining academic rigor

- Technical Problem-Solving: I developed systematic approaches for troubleshooting complex integration challenges
- Academic Writing and Documentation: I created comprehensive documentation suitable for both academic publication and technical reproduction

I found the following topics particularly challenging:

Container networking and GUI integration proved most challenging, requiring deep investigation of Docker networking, Qt platform dependencies, and Linux display systems. These challenges taught me valuable lesson about systematic troubleshooting and the importance of understanding underlying system architecture. I learned that complex technical problems often have elegant solutions once proper understanding is achieved.

I found the following aspects particularly interesting:

The intersection of cybersecurity and autonomous systems fascinated me, particularly how containerized approaches enable reproducible research while addressing real-world security threats. I was intrigued by the potential for machine learning to solve detection challenges that traditional approaches cannot handle effectively.

I still need to work on the following areas:

- Advanced Machine Learning Implementation: Developing practical experience with LSTM networks and DTW algorithms for my SIT724 continuation
- Hardware Deployment: Transitioning from simulation to embedded hardware platforms
- Publication Process: Learning academic conference and journal submission processes
- Industry Collaboration: Developing connections with autonomous vehicle industry partners

The things that helped me most were:

- Systematic Documentation: Maintaining detailed implementation logs enabled effective troubleshooting
- Supervisor Guidance: Dr. Maiti's expertise provided crucial direction during challenging phases
- Technical Communities: Docker, PX4, and ROS communities offered invaluable support
- Iterative Development: Breaking complex implementation into manageable phases ensured steady progress

If I did the research unit again, I would do the following things differently:

I would allocate more initial time for infrastructure setup and engage earlier with technical communities to leverage existing knowledge. I would also document solutions more comprehensively to contribute back to the research community more effectively.

Other...:

This unit transformed my understanding of research from theoretical knowledge to practical implementation capability. I developed confidence in autonomous problem-solving while

maintaining rigorous methodology. The experience prepared me for independent research continuation in SIT724 and potential PhD studies. I now understand that meaningful research contributions emerge from addressing real-world challenges through systematic investigation combined with technical excellence.

The containerized architecture I developed demonstrates capability to bridge academic research with industry-applicable solutions, positioning me for continued success in autonomous systems cybersecurity research while contributing valuable tools to the broader research community.

DEAKIN UNIVERSITY

RESEARCH TRAINING AND PROJECT

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Portfolio Submission

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October 2, 2025



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